Djordje Miladinovic

Curriculum Vitæ

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Summary

- **Profile**: Senior AI/ML researcher & engineer | PhD in Computer Science (AI/ML) | Swiss resident | Serbian citizen. 10+ years in AI/ML research | 15+ years in software development | 5+ years in project leading | 5+ years in life sciences.
- **Work**: Designing and leading the end-to-end development of scalable AI/ML and software products, from concept to deployment.
- **Mission**: My goal is to utilize technology, innovative design, and data-driven insights to improve individual and societal well-being, fuel business growth, and deliver impactful, best-in-class solutions.

Experience

Senior AI/ML Engineer, *AI department of GSK plc* — ***Oct'21-Present***

- Spearheaded the development of a foundational model integrating omics data to accelerate preclinical drug discovery, enabling target identification and uncovering disease and drug mechanisms – *adopted by three R&D units at GSK, enhancing decision-making and aiding in the identification of optimal candidate protein targets*.
- Supported clinical efforts to optimize the use of existing GSK drugs by applying survival analysis and processing large-scale electronic health record (EHR) databases – *to be used by clinicians for improved diagnosis and prognosis*.

Machine Learning Researcher, *ETH Zürich* — ***Feb'17-Jun'21***

- See the full list of my publications at Google Scholar. Notable projects include:
- Sleep-learning web platform: Led a team of researchers to develop a web-based large-scale machine learning platform for sleep pattern recognition from EEG/EMG signals – *now commercialized with over 100K jobs processed*.
- SDN neural networks: Co-developed a neural architecture for realistic image synthesis – *state-of-the-art performance*.
- Granger-causality framework for multimodal analysis of mass spectrometry and EEG data: Co-developed an explainable AI (XAI) algorithm to elucidate causal relationships between metabolism and sleep.

Machine Learning Researcher (Visitor), *Max Planck Institute for Intelligent Systems* — ***Sep'18-Dec'18***

- Co-developed a framework for evaluating disentangled representations – *has about 200 scientific citations so far*.

Machine Learning Researcher & Engineer (Intern), *Logitech Europe S.A.* — ***Sep'16-Feb'17***

- Developed an AI/ML product to predict potential premium users and reduce churn rate.

Machine Learning Researcher (Visitor), *Disney Research Studios, Zurich* — ***Jan'16-Sep'16***

- Developed a software platform for detecting mechanical failures in Disney's humanoid robots based on IMU sensor readings. The platform uses novel machine learning algorithm to asses robot degradation in a human-like way.

Education

PhD in Computer Science – Machine Learning, *ETH Zürich* — ***Sep'17-Jun'21***

- *Thesis:* "On training Deep Generative Models with Latent Variables" – Prof. Joachim M. Buhmann.
- *Research areas:* Generative image and text modeling | variational autoencoders | representation learning | deep learning | language models | machine learning for biology | drug discovery;

MS in Computer Science – Machine Learning, *ETH Zürich* — ***Sep'13-Jun'16***

- *Thesis:* "Perceptual Analysis Framework for Discovering Anomalies in Humanoid Motions" – Prof. Otmar Hilliges.
- *Focus:* Machine learning | software engineering | operating systems | distributed computing;

BS in Computer Science – Software Engineering, *ETF Belgrade* — ***Sep'09-Sep'13***

- *Thesis:* "Java-based Interactive Software Simulator of a CISC Processor" – Prof. Zaharije Radivojević.
- *Focus:* Software engineering and design | algorithms | data structures | operating systems | hardware design;

Technical Skills

Programming	*Advanced:* Python (PyTorch);
	Good but rusty: C/C++ | Java | Javascript | Bash | SQL;
Infrastructure	*Data formats:* Parquet | H5 | H5AD | CSV | YAML | JSON | Pickle;
	Databases: PostgreSQL | MongoDB | SQLite | Milvus;
	Big data: Spark | Hive | Hadoop;
	AI/ML frameworks: Pytorch Lightning | Ray | Weights & Biases | MLflow | HuggingFace | Kubeflow;
	High Performance Computing (HPC): SLURM | LSF | Condor;
	Middleware: Celery | Redis | Kafka;
	Cloud: GCP/GCS | AWS/S3 | Microsoft Azure;
	Deployment: Docker | Kubernetes;
	CI/CD: GitHub Actions | CircleCI;
	Web: Node.js | Django | HTML/CSS;
Python libraries	*Data Science:* Pandas | SciPy | Seaborn | NumPy | Scanpy | Scrapy | OpenCV;
	AI/ML: PyTorch | Scikit-learn | SpaCy | Fairseq | LangChain | Tensorflow;
Design tools	Figma | Inkscape | Blender | PowerPoint | Adobe Illustrator;

Languages

Serbian (Native) | English (Fluent) | German (Intermediate) | Spanish (Beginner);

Hobbies

Waterpolo | Swimming | Basketball | Tennis | Chess | Cinematography | Reading;